

ADDITIONAL MATERIAL FOR: A Post-Processing Heuristic for the Static Separation of Permissions Problem

Carlo Blundo
DISA-MIS
Università di Salerno, Italy

Stelvio Cimato
Dipartimento di Informatica
Università di Milano, Italy

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1 Real World Datasets

The datasets listed in Table 1 were sourced from HP Labs and were first used for evaluation in [1]. For eight of the datasets, the optimal unconstrained solutions are known and reported in [1]. Table 1 provides key parameters.

Dataset	$ \mathcal{U} $	$ \mathcal{P} $	$ \mathcal{UPA} $	$ \mathcal{R} $	Density
Americas large	3485	10127	185294	398	0.00525
Americas small	3477	1587	105205	178	0.01907
Apj	2044	1164	6841	453	0.00288
Customer	10021	277	45427	//	0.01637
Domino	79	231	730	20	0.04000
Emea	35	3046	7220	34	0.06772
Firewall 1	365	709	31951	66	0.12347
Firewall 2	325	590	36428	10	0.18998
Healthcare	46	46	1486	14	0.70227

Table 1: Real world datasets

These datasets represent a variety of access control scenarios:

- **Network Access Rules:** Datasets like Americas small, Americas large, Apj, and Emea were derived from Hewlett-Packard’s (HP) network access control rules. The Americas profiles, for instance, were extracted from Cisco firewalls that manage access for external business partners.
- **Firewall Analysis:** The Firewall 1 and Firewall 2 datasets were created by analyzing Checkpoint firewalls to determine if service packets could reach specific network destinations.
- **Healthcare Permissions:** The Healthcare dataset was provided by the U.S. Department of Veterans Affairs and contains a list of permissions for licensed or certified healthcare providers.
- **Server Access:** The Domino dataset is based on user and access profiles for a Lotus Domino server.
- **Customer-Specific Data:** The Customer dataset originates from an access control graph obtained in collaboration with the IT department of an HP customer. For this dataset, the optimal solution is not known.

In our experiments, during the $\mathcal{CP}$ generation phase, we set $nc = |\mathcal{CP}| \leq |\mathcal{U}|/10$, for the modalities **upa-***, and $nc = |\mathcal{CP}| \leq |\mathcal{R}|/10$, for the modalities **pa-***. We refer to such a setting as the *proportional setting*. We also considered a *constant setting*, where we set $nc = 8$.

1.1 Americas large – nc = 8

V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	415	93255	0.9137	0.3921
OI	413	93070	0.9163	0.3983
UL	415	93294	0.9101	0.3641
UI	415	93206	0.9084	0.385

Table 2: Starting decomposition – Dataset: Americas large

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	420	96116	0.908	0.3771	0.9939	0.9647
	OI	419	95994	0.9095	0.3801	0.9927	0.9576
	UL	420	96155	0.9045	0.3498	0.9939	0.9647
	UI	421	96130	0.9019	0.3673	0.9929	0.9578
upa-small	OL	422	93309	0.9043	0.3689	0.9899	0.9465
	OI	421	93126	0.9059	0.3742	0.9891	0.9441
	UL	423	93350	0.8999	0.3415	0.9889	0.9443
	UI	423	93262	0.8981	0.3615	0.9888	0.9443
pa-rnd	OL	421	93269	0.9071	0.3742	0.9927	0.9578
	OI	419	93084	0.9097	0.3801	0.9927	0.9576
	UL	421	93308	0.9035	0.347	0.9927	0.9578
	UI	421	93220	0.9018	0.3673	0.9927	0.9578
pa-small	OL	422	93291	0.9028	0.3689	0.9878	0.933
	OI	420	93106	0.9054	0.3748	0.9877	0.9327
	UL	423	93330	0.8998	0.3437	0.9879	0.922
	UI	423	93242	0.8981	0.3638	0.9879	0.922

Table 3: Post-processing decomposition – Dataset: Americas large

1.2 Americas small – nc = 8

V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	196	11200	0.7497	0.2343
OI	196	11164	0.7493	0.2303
UL	212	10862	0.7141	0.1927
UI	214	10681	0.7182	0.1915

Table 4: Starting decomposition – Dataset: Americas small

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	197	14058	0.7468	0.2295	0.9973	0.9848
	OI	197	14022	0.7464	0.2255	0.9973	0.9848
	UL	213	13720	0.7114	0.1884	0.9975	0.986
	UI	215	13539	0.7156	0.1873	0.9976	0.9861
upa-small	OL	202	11513	0.7335	0.2063	0.9799	0.9135
	OI	202	11477	0.7331	0.2025	0.9799	0.9135
	UL	218	11175	0.6994	0.1681	0.9817	0.9196
	UI	219	10982	0.7052	0.1711	0.9839	0.933
pa-rnd	OL	202	11382	0.7392	0.2141	0.9856	0.9135
	OI	202	11346	0.7387	0.2102	0.9855	0.9135
	UL	216	11036	0.7072	0.1796	0.9904	0.9455
	UI	217	10853	0.7128	0.1826	0.9926	0.9591
pa-small	OL	203	11397	0.7278	0.2019	0.9738	0.9
	OI	203	11361	0.7274	0.1981	0.9738	0.9
	UL	219	11059	0.694	0.1642	0.9758	0.9071
	UI	221	10878	0.6982	0.1633	0.976	0.9079

Table 5: Post-processing decomposition – Dataset: Americas small

1.3 Apj – nc = 8

V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	454	5160	0.8883	0.5478
OI	454	5148	0.8894	0.5504
UL	455	5045	0.8871	0.5416
UI	455	5043	0.8881	0.5442

Table 6: Starting decomposition – Dataset: Apj

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	458	5254	0.8815	0.5337	0.9936	0.974
	OI	458	5242	0.8826	0.5363	0.9936	0.974
	UL	459	5139	0.8803	0.5276	0.9936	0.9741
	UI	459	5137	0.8814	0.5302	0.9936	0.9741
upa-small	OL	459	5319	0.8804	0.5302	0.9918	0.9677
	OI	459	5307	0.8815	0.5328	0.9918	0.9677
	UL	460	5204	0.8792	0.5242	0.9918	0.9677
	UI	460	5202	0.8803	0.5268	0.9918	0.9677
pa-rnd	OL	458	5173	0.8828	0.5337	0.9943	0.974
	OI	458	5161	0.8839	0.5363	0.9943	0.974
	UL	459	5058	0.8816	0.5276	0.994	0.9741
	UI	459	5057	0.8815	0.5276	0.9931	0.9698
pa-small	OL	459	5319	0.8804	0.5302	0.9918	0.9677
	OI	459	5307	0.8815	0.5328	0.9918	0.9677
	UL	460	5204	0.8792	0.5242	0.9918	0.9677
	UI	460	5202	0.8803	0.5268	0.9918	0.9677

Table 7: Post-processing decomposition – Dataset: Apj

1.4 Domino – nc = 8

V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	20	747	0.9294	0.4286
OI	20	761	0.9017	0.3333
UL	20	761	0.9017	0.3333
UI	20	761	0.9017	0.3333

Table 8: Starting decomposition – Dataset: Domino

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	20	752	0.9294	0.4286	0.95	0.8182
	OI	20	761	0.9017	0.3333	1.0	1.0
	UL	20	761	0.9017	0.3333	1.0	1.0
	UI	20	761	0.9017	0.3333	1.0	1.0
upa-small	OL	20	761	0.9128	0.3793	0.9333	0.7391
	OI	20	761	0.9017	0.3333	1.0	1.0
	UL	20	761	0.9017	0.3333	1.0	1.0
	UI	20	761	0.9017	0.3333	1.0	1.0
pa-rnd	OL	22	753	0.8972	0.3548	0.9411	0.75
	OI	20	761	0.9017	0.3333	1.0	1.0
	UL	20	761	0.9017	0.3333	1.0	1.0
	UI	20	761	0.9017	0.3333	1.0	1.0
pa-small	OL	22	762	0.8813	0.3125	0.9247	0.68
	OI	20	761	0.9017	0.3333	1.0	1.0
	UL	20	761	0.9017	0.3333	1.0	1.0
	UI	20	761	0.9017	0.3333	1.0	1.0

Table 9: Post-processing decomposition – Dataset: Domino

1.5 Emea – nc = 8

V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	34	7280	1.0	1.0
OI	34	7280	1.0	1.0
UL	34	7280	1.0	1.0
UI	34	7280	1.0	1.0

Table 10: Starting decomposition – Dataset: Emea

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	42	7296	0.9048	0.52	0.9048	0.52
	OI	42	7296	0.9046	0.52	0.9046	0.52
	UL	42	7296	0.9047	0.52	0.9047	0.52
	UI	42	7296	0.9046	0.52	0.9046	0.52
upa-small	OL	42	7297	0.8993	0.52	0.8993	0.52
	OI	42	7297	0.9002	0.52	0.9002	0.52
	UL	41	7295	0.9078	0.5306	0.9078	0.5306
	UI	42	7297	0.8993	0.52	0.8993	0.52
pa-rnd	OL	42	7296	0.9044	0.52	0.9044	0.52
	OI	42	7296	0.9033	0.52	0.9033	0.52
	UL	42	7296	0.9038	0.52	0.9038	0.52
	UI	42	7296	0.9035	0.52	0.9035	0.52
pa-small	OL	41	7295	0.9093	0.5306	0.9093	0.5306
	OI	41	7295	0.9094	0.5306	0.9094	0.5306
	UL	41	7295	0.908	0.5306	0.908	0.5306
	UI	42	7297	0.899	0.52	0.899	0.52

Table 11: Post-processing decomposition – Dataset: Emea

1.6 Firewall 1 – nc = 8

V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	65	3294	0.7999	0.3646
OI	65	3275	0.7955	0.3367
UL	69	3248	0.785	0.3366
UI	69	3236	0.7853	0.3366

Table 12: Starting decomposition – Dataset: Firewall 1

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	68	3318	0.7762	0.3137	0.9712	0.8732
	OI	67	3297	0.7793	0.3039	0.9803	0.913
	UL	71	3265	0.7697	0.3048	0.9814	0.9178
	UI	71	3253	0.77	0.3048	0.9814	0.9178
upa-small	OL	71	3620	0.7064	0.2455	0.9212	0.7215
	OI	71	3593	0.7096	0.2342	0.9286	0.7436
	UL	75	3612	0.6947	0.2261	0.9256	0.7349
	UI	75	3593	0.7021	0.2368	0.9325	0.7561
pa-rnd	OL	67	3550	0.7881	0.3434	0.9849	0.913
	OI	66	3526	0.7895	0.32	0.9923	0.9552
	UL	70	3499	0.7794	0.3204	0.9927	0.9577
	UI	70	3487	0.7797	0.3204	0.9927	0.9577
pa-small	OL	71	3591	0.7052	0.2455	0.9199	0.7215
	OI	71	3564	0.7084	0.2342	0.9273	0.7436
	UL	75	3583	0.6934	0.2261	0.9243	0.7349
	UI	75	3564	0.7008	0.2368	0.9313	0.7561

Table 13: Post-processing decomposition – Dataset: Firewall 1

1.7 Firewall 2 – nc = 8

V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	10	1542	0.7655	0.25
OI	10	1564	0.8259	0.25
UL	10	1787	0.7877	0.25
UI	10	1787	0.7877	0.25

Table 14: Starting decomposition – Dataset: Firewall 2

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	11	1828	0.7291	0.1667	0.9516	0.75
	OI	11	1850	0.7882	0.1667	0.9516	0.75
	UL	11	2073	0.7516	0.1667	0.9516	0.75
	UI	11	2073	0.7516	0.1667	0.9516	0.75
upa-small	OL	12	1918	0.6866	0.1	0.9054	0.5714
	OI	12	1940	0.7446	0.1	0.9054	0.5714
	UL	12	2163	0.7094	0.1	0.9054	0.5714
	UI	12	2163	0.7094	0.1	0.9054	0.5714
pa-rnd	OL	11	1828	0.7291	0.1667	0.9516	0.75
	OI	11	1850	0.7882	0.1667	0.9516	0.75
	UL	11	2073	0.7516	0.1667	0.9516	0.75
	UI	11	2073	0.7516	0.1667	0.9516	0.75
pa-small	OL	14	2072	0.5927	0.0	0.7959	0.3333
	OI	14	2094	0.6489	0.0	0.7959	0.3333
	UL	14	2317	0.6159	0.0	0.7959	0.3333
	UI	14	2317	0.6159	0.0	0.7959	0.3333

Table 15: Post-processing decomposition – Dataset: Firewall 2

1.8 Healthcare – nc = 8

V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	14	385	0.7594	0.4
OI	14	384	0.732	0.3333
UL	14	306	0.7644	0.4
UI	14	351	0.7469	0.3333

Table 16: Starting decomposition – Dataset: Healthcare

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	15	404	0.7422	0.3182	0.9735	0.8125
	OI	14	384	0.732	0.3333	1.0	1.0
	UL	15	325	0.735	0.3182	0.9624	0.8125
	UI	14	351	0.7469	0.3333	1.0	1.0
upa-small	OL	16	450	0.716	0.25	0.9419	0.6667
	OI	15	430	0.7047	0.2609	0.965	0.8125
	UL	16	371	0.7204	0.25	0.9419	0.6667
	UI	15	397	0.7189	0.2609	0.965	0.8125
pa-rnd	OL	16	438	0.6691	0.2	0.8922	0.5789
	OI	16	438	0.6535	0.2	0.9018	0.6667
	UL	16	359	0.6735	0.2	0.8922	0.5789
	UI	16	405	0.667	0.2	0.9018	0.6667
pa-small	OL	20	559	0.5491	0.0303	0.7503	0.2593
	OI	20	559	0.5355	0.0303	0.759	0.3077
	UL	20	446	0.5526	0.0303	0.7503	0.2593
	UI	20	526	0.5471	0.0303	0.759	0.3077

Table 17: Post-processing decomposition – Dataset: Healthcare

1.9 Americas large – nc = 10%

	V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	415	93255	0.9137	0.3921	
OI	413	93070	0.9163	0.3983	
UL	415	93294	0.9101	0.3641	
UI	415	93206	0.9084	0.385	

Table 18: Starting decomposition – Dataset: Americas large

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	590	96615	0.7782	0.0774	0.849	0.2934
	OI	597	96510	0.7753	0.078	0.8433	0.2883
	UL	565	96571	0.7896	0.082	0.8655	0.3573
	UI	575	96561	0.7822	0.0811	0.8592	0.3378
upa-small	OL	602	96683	0.771	0.0627	0.8415	0.2681
	OI	607	96574	0.7691	0.0635	0.8365	0.2639
	UL	580	96644	0.7818	0.0677	0.8546	0.3267
	UI	586	96626	0.7752	0.0661	0.8511	0.3102
pa-rnd	OL	442	96141	0.8847	0.3187	0.9676	0.8118
	OI	440	95956	0.8875	0.3239	0.9674	0.811
	UL	442	96177	0.8832	0.2963	0.9692	0.808
	UI	444	96097	0.8794	0.3115	0.9668	0.7971
pa-small	OL	493	96453	0.8396	0.2638	0.9152	0.5416
	OI	494	96279	0.8385	0.2652	0.9113	0.5347
	UL	473	96429	0.8534	0.2623	0.9331	0.6264
	UI	479	96350	0.8454	0.2601	0.9289	0.5879

Table 19: Post-processing decomposition – Dataset: Americas large

1.10 Americas small – nc = 10%

	V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	196	11200	0.7497	0.2343	
OI	196	11164	0.7493	0.2303	
UL	212	10862	0.7141	0.1927	
UI	214	10681	0.7182	0.1915	

Table 20: Starting decomposition – Dataset: Americas small

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	202	14106	0.7364	0.2102	0.9834	0.9135
	OI	202	14070	0.7366	0.2063	0.9839	0.9135
	UL	214	13739	0.7086	0.1843	0.994	0.9722
	UI	216	13558	0.7127	0.1832	0.9941	0.9725
upa-small	OL	212	14886	0.716	0.1747	0.9565	0.7895
	OI	212	14850	0.7162	0.1677	0.9571	0.7895
	UL	222	14254	0.6923	0.1527	0.9722	0.8707
	UI	223	14061	0.6983	0.1556	0.9752	0.8836
pa-rnd	OL	204	11342	0.7325	0.205	0.9771	0.8868
	OI	204	11306	0.732	0.2013	0.9774	0.8868
	UL	218	10995	0.7009	0.1716	0.9832	0.9196
	UI	220	10814	0.7052	0.1706	0.9831	0.9204
pa-small	OL	214	11586	0.707	0.1701	0.9431	0.7447
	OI	214	11603	0.7069	0.1701	0.9421	0.7447
	UL	228	11224	0.6783	0.1404	0.9517	0.7959
	UI	229	11046	0.6804	0.1369	0.9533	0.8082

Table 21: Post-processing decomposition – Dataset: Americas small

1.11 Apj – nc = 10%

V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	454	5160	0.8883	0.5478
OI	454	5148	0.8894	0.5504
UL	455	5045	0.8871	0.5416
UI	455	5043	0.8881	0.5442

Table 22: Starting decomposition – Dataset: Apj

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	522	5526	0.8044	0.3504	0.9091	0.6542
	OI	522	5514	0.8056	0.3523	0.9096	0.6542
	UL	521	5407	0.8047	0.3472	0.9111	0.657
	UI	522	5407	0.8052	0.3485	0.9113	0.6559
upa-small	OL	576	6166	0.7373	0.2279	0.8396	0.4589
	OI	579	6157	0.7374	0.2286	0.8386	0.457
	UL	578	6050	0.7365	0.2259	0.8407	0.4611
	UI	579	6048	0.7369	0.2271	0.84	0.4605
pa-rnd	OL	477	5250	0.8583	0.4692	0.9675	0.8583
	OI	478	5240	0.8587	0.4708	0.9664	0.8566
	UL	476	5128	0.8596	0.4699	0.9705	0.8695
	UI	476	5126	0.8605	0.4723	0.9703	0.8695
pa-small	OL	487	5582	0.8381	0.4373	0.9465	0.8027
	OI	487	5570	0.8392	0.4395	0.9465	0.8027
	UL	487	5466	0.8377	0.4351	0.9474	0.8081
	UI	487	5465	0.8376	0.4351	0.9464	0.8046

Table 23: Post-processing decomposition – Dataset: Apj

1.12 Domino – nc = 10%

V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	20	747	0.9294	0.4286
OI	20	761	0.9017	0.3333
UL	20	761	0.9017	0.3333
UI	20	761	0.9017	0.3333

Table 24: Starting decomposition – Dataset: Domino

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	20	752	0.9294	0.4286	0.95	0.8182
	OI	20	761	0.9017	0.3333	1.0	1.0
	UL	20	761	0.9017	0.3333	1.0	1.0
	UI	20	761	0.9017	0.3333	1.0	1.0
upa-small	OL	21	763	0.899	0.3667	0.919	0.7083
	OI	20	761	0.9017	0.3333	1.0	1.0
	UL	20	761	0.9017	0.3333	1.0	1.0
	UI	20	761	0.9017	0.3333	1.0	1.0
pa-rnd	OL	20	747	0.9294	0.4286	1.0	1.0
	OI	20	761	0.9017	0.3333	1.0	1.0
	UL	20	761	0.9017	0.3333	1.0	1.0
	UI	20	761	0.9017	0.3333	1.0	1.0
pa-small	OL	22	762	0.8902	0.3548	0.9337	0.75
	OI	20	761	0.9017	0.3333	1.0	1.0
	UL	20	761	0.9017	0.3333	1.0	1.0
	UI	20	761	0.9017	0.3333	1.0	1.0

Table 25: Post-processing decomposition – Dataset: Domino

1.13 Emea – nc = 10%

V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	34	7280	1.0	1.0
OI	34	7280	1.0	1.0
UL	34	7280	1.0	1.0
UI	34	7280	1.0	1.0

Table 26: Starting decomposition – Dataset: Emea

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	37	7286	0.9594	0.775	0.9594	0.775
	OI	37	7286	0.9594	0.775	0.9594	0.775
	UL	37	7286	0.9596	0.775	0.9596	0.775
	UI	37	7286	0.9594	0.775	0.9594	0.775
upa-small	OL	36	7285	0.9673	0.7949	0.9673	0.7949
	OI	37	7287	0.9555	0.775	0.9555	0.775
	UL	37	7287	0.9555	0.775	0.9555	0.775
	UI	37	7287	0.9559	0.775	0.9559	0.775
pa-rnd	OL	37	7286	0.9593	0.775	0.9593	0.775
	OI	37	7286	0.9592	0.775	0.9592	0.775
	UL	37	7286	0.9592	0.775	0.9592	0.775
	UI	37	7286	0.9593	0.775	0.9593	0.775
pa-small	OL	36	7285	0.9677	0.7949	0.9677	0.7949
	OI	37	7287	0.9561	0.775	0.9561	0.775
	UL	36	7285	0.9673	0.7949	0.9673	0.7949
	UI	37	7287	0.9559	0.775	0.9559	0.775

Table 27: Post-processing decomposition – Dataset: Emea

1.14 Firewall 1 – nc = 10%

V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	65	3294	0.7999	0.3646
OI	65	3275	0.7955	0.3367
UL	69	3248	0.785	0.3366
UI	69	3236	0.7853	0.3366

Table 28: Starting decomposition – Dataset: Firewall 1

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	76	3581	0.7182	0.2034	0.9005	0.6023
	OI	76	3562	0.726	0.2034	0.9041	0.6207
	UL	80	3524	0.7136	0.1967	0.9114	0.6374
	UI	80	3512	0.7129	0.1967	0.9101	0.6374
upa-small	OL	85	4384	0.638	0.1103	0.8269	0.4019
	OI	84	4355	0.646	0.1111	0.8401	0.4327
	UL	87	4361	0.637	0.1087	0.8447	0.4444
	UI	85	4337	0.6415	0.1103	0.8551	0.4667
pa-rnd	OL	68	3302	0.7779	0.3137	0.973	0.8732
	OI	68	3283	0.7737	0.2885	0.973	0.8732
	UL	72	3255	0.7645	0.2897	0.9745	0.88
	UI	72	3244	0.7648	0.2897	0.9745	0.88
pa-small	OL	69	3567	0.7228	0.2736	0.9407	0.7867
	OI	69	3540	0.7261	0.2617	0.9482	0.8108
	UL	73	3559	0.7103	0.2523	0.944	0.7975
	UI	73	3540	0.7177	0.2636	0.9511	0.8205

Table 29: Post-processing decomposition – Dataset: Firewall 1

1.15 Firewall 2 – nc = 10%

V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	10	1542	0.7655	0.25
OI	10	1564	0.8259	0.25
UL	10	1787	0.7877	0.25
UI	10	1787	0.7877	0.25

Table 30: Starting decomposition – Dataset: Firewall 2

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	11	1828	0.7291	0.1667	0.9516	0.75
	OI	11	1850	0.7882	0.1667	0.9516	0.75
	UL	11	2073	0.7516	0.1667	0.9516	0.75
	UI	11	2073	0.7516	0.1667	0.9516	0.75
upa-small	OL	12	1918	0.6866	0.1	0.9054	0.5714
	OI	12	1940	0.7446	0.1	0.9054	0.5714
	UL	12	2163	0.7094	0.1	0.9054	0.5714
	UI	12	2163	0.7094	0.1	0.9054	0.5714
pa-rnd	OL	11	1643	0.707	0.1667	0.9295	0.75
	OI	11	1665	0.7661	0.1667	0.9295	0.75
	UL	11	1888	0.7296	0.1667	0.9295	0.75
	UI	11	1888	0.7296	0.1667	0.9295	0.75
pa-small	OL	11	1595	0.707	0.1667	0.9295	0.75
	OI	11	1617	0.7661	0.1667	0.9295	0.75
	UL	11	1840	0.7296	0.1667	0.9295	0.75
	UI	11	1840	0.7296	0.1667	0.9295	0.75

Table 31: Post-processing decomposition – Dataset: Firewall 2

1.16 Healthcare – nc = 10%

V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	14	385	0.7594	0.4
OI	14	384	0.732	0.3333
UL	14	306	0.7644	0.4
UI	14	351	0.7469	0.3333

Table 32: Starting decomposition – Dataset: Healthcare

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	15	404	0.7303	0.3182	0.9624	0.8125
	OI	14	384	0.732	0.3333	1.0	1.0
	UL	15	325	0.7469	0.3182	0.9735	0.8125
	UI	14	351	0.7469	0.3333	1.0	1.0
upa-small	OL	16	450	0.716	0.25	0.9419	0.6667
	OI	15	430	0.7047	0.2609	0.965	0.8125
	UL	16	371	0.7204	0.25	0.9419	0.6667
	UI	15	397	0.7189	0.2609	0.965	0.8125
pa-rnd	OL	14	385	0.7594	0.4	1.0	1.0
	OI	14	384	0.732	0.3333	1.0	1.0
	UL	14	306	0.7644	0.4	1.0	1.0
	UI	14	351	0.7469	0.3333	1.0	1.0
pa-small	OL	15	414	0.7175	0.3182	0.9488	0.8125
	OI	15	413	0.6913	0.2609	0.9488	0.8125
	UL	15	318	0.7222	0.3182	0.9488	0.8125
	UI	15	380	0.7055	0.2609	0.9488	0.8125

Table 33: Post-processing decomposition – Dataset: Healthcare

2 Synthetic Datasets

To test the heuristic, we also resort to synthetic datasets. The random data generator, first introduced in [2], uses five parameters to simulate various access control configurations: the number of users nu ; the number of roles nr ; the number of permissions np ; the maximum number of roles a user can have mru ; the maximum number of permissions a role can have mpr . The generation process involves three main steps: *Role-Permission Assignment*, for each role, a random number of permissions (up to mpr) are randomly selected and assigned; *User-Role Assignment*, each user is assigned a random number of roles (up to mru); *User-Permission Computation*: finally, user-permission assignments are calculated based on the established user-role and role-permission relationships. It's important to note that the data produced by this generator is unstructured, treating each user, role, and permission as statistically independent entities.

2.1 Varying number of permissions

Dataset	nr	nu	np	$mrcu$	mpr
d11	100	2000	100	3	10
d12	100	2000	500	3	50
d13	100	2000	1000	3	100
d14	100	2000	2000	3	200

Table 34: Constant nr and nu , varying np , and $mrcu \times mpr \approx np/3.33$

2.1.1 Dataset d11 – $nc = 8$

V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	100	7435	0.8945	0.3423
OI	100	7435	0.8945	0.3423
UL	159	18545	0.4647	0.0975
UI	147	19063	0.4531	0.1027

Table 35: Starting decomposition – Dataset: d11

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	100	7435	0.8945	0.3423	1.0	1.0
	OI	100	7435	0.8945	0.3423	1.0	1.0
	UL	159	18545	0.4647	0.0975	1.0	1.0
	UI	147	19063	0.4531	0.1027	1.0	1.0
upa-small	OL	105	8114	0.8279	0.2654	0.9637	0.8636
	OI	105	8114	0.8279	0.2654	0.9637	0.8636
	UL	143	17857	0.4401	0.0848	0.9897	0.9595
	UI	141	18191	0.4204	0.0711	0.9931	0.9724
pa-rnd	OL	101	9013	0.7999	0.1824	0.985	0.9417
	OI	101	9014	0.7999	0.1824	0.985	0.9417
	UL	137	19716	0.3978	0.058	0.9964	0.9856
	UI	140	19571	0.3888	0.0526	0.9965	0.9859
pa-small	OL	107	7560	0.8414	0.3101	0.9498	0.8158
	OI	107	7560	0.8414	0.3101	0.9498	0.8158
	UL	156	18081	0.416	0.0579	0.9893	0.9571
	UI	145	18326	0.4016	0.0515	0.9885	0.9539

Table 36: Post-processing decomposition – Dataset: d11

2.1.2 Dataset d12 – nc = 8

	V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	100	6890	6890	0.9936	0.8349
OI	100	6890	6890	0.9936	0.8349
UL	922	37783	37783	0.7163	0.022
UI	920	37413	37413	0.6872	0.022

Table 37: Starting decomposition – Dataset: d12

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	105	7071	0.9675	0.7227	0.9738	0.8636
	OI	105	7071	0.9672	0.7227	0.9735	0.8636
	UL	923	37853	0.7136	0.0199	0.9991	0.9946
	UI	922	37485	0.6844	0.02	0.9987	0.9935
upa-small	OL	106	7520	0.9534	0.5969	0.9613	0.8393
	OI	106	7520	0.9534	0.5969	0.9613	0.8393
	UL	918	36749	0.6894	0.0231	0.9964	0.9827
	UI	922	35765	0.6872	0.021	0.9966	0.9828
pa-rnd	OL	108	7198	0.9591	0.6774	0.9619	0.7931
	OI	108	7198	0.9591	0.6774	0.9618	0.7931
	UL	915	36332	0.7124	0.0263	1.0	1.0
	UI	929	35541	0.6897	0.0259	1.0	1.0
pa-small	OL	108	7580	0.9443	0.6124	0.9492	0.7931
	OI	108	7580	0.9443	0.6124	0.9492	0.7931
	UL	941	39057	0.69	0.0206	0.9957	0.9789
	UI	935	39148	0.69	0.0167	0.9962	0.9809

Table 38: Post-processing decomposition – Dataset: d12

2.1.3 Dataset d13 – nc = 8

	V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	100	100	9659	0.998	0.8349
OI	100	100	9659	0.998	0.8349
UL	1195	1195	85554	0.794	0.0157
UI	1195	1195	86084	0.7956	0.0157

Table 39: Starting decomposition – Dataset: d13

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	102	9737	0.9882	0.7876	0.9902	0.9423
	OI	102	9737	0.9882	0.7876	0.9901	0.9423
	UL	1195	85554	0.794	0.0157	1.0	1.0
	UI	1195	86084	0.7956	0.0157	1.0	1.0
upa-small	OL	108	9785	0.9511	0.7778	0.9514	0.7931
	OI	108	9785	0.9511	0.7778	0.9514	0.7931
	UL	1153	86538	0.7873	0.0072	0.9958	0.9802
	UI	1150	86973	0.7848	0.0073	0.9963	0.981
pa-rnd	OL	108	9671	0.9617	0.7049	0.9625	0.7931
	OI	108	9671	0.9618	0.7049	0.9627	0.7931
	UL	1173	85582	0.7754	0.0119	1.0	1.0
	UI	1173	84722	0.7693	0.0119	1.0	1.0
pa-small	OL	108	9531	0.9501	0.7778	0.9504	0.7931
	OI	108	9531	0.9501	0.7778	0.9504	0.7931
	UL	1113	81863	0.7821	0.0083	0.9961	0.9813
	UI	1111	81650	0.7775	0.0083	0.9963	0.9821

Table 40: Post-processing decomposition – Dataset: d13

2.1.4 Dataset d14 – nc = 8

	V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	100	100	14206	0.9999	0.9802
OI	100	100	14206	0.9999	0.9802
UL	1236	1236	184920	0.8321	0.0137
UI	1237	1237	185429	0.8318	0.0136

Table 41: Starting decomposition – Dataset: d14

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	105	14411	0.9758	0.8468	0.9758	0.8636
	OI	105	14411	0.9759	0.8468	0.9759	0.8636
	UL	1237	184957	0.8316	0.0129	0.9996	0.9976
	UI	1238	185466	0.8313	0.0129	0.9996	0.9976
upa-small	OL	108	14177	0.9519	0.7778	0.952	0.7931
	OI	108	14177	0.9519	0.7778	0.952	0.7931
	UL	1204	176417	0.8264	0.0116	0.9966	0.9818
	UI	1208	176427	0.8254	0.0116	0.9959	0.9803
pa-rnd	OL	108	14002	0.9617	0.7931	0.9617	0.7931
	OI	108	14002	0.9618	0.7931	0.9618	0.7931
	UL	1258	179893	0.8276	0.0112	0.9995	0.9976
	UI	1258	178799	0.8249	0.0112	0.9995	0.9976
pa-small	OL	108	13424	0.9566	0.7931	0.9566	0.7931
	OI	108	13424	0.9566	0.7931	0.9566	0.7931
	UL	1221	163047	0.8264	0.0084	0.9964	0.9805
	UI	1221	165985	0.8262	0.0076	0.9967	0.9813

Table 42: Post-processing decomposition – Dataset: d14

2.1.5 Dataset d11 – nc = 10%

	V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	100	7971	7971	0.8359	0.2422
OI	100	7971	7971	0.8359	0.2422
UL	169	16663	16663	0.471	0.0935
UI	172	16944	16944	0.4575	0.0837

Table 43: Starting decomposition – Dataset: d11

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	104	8201	0.8143	0.2071	0.9773	0.8889
	OI	104	8201	0.8139	0.2071	0.9761	0.8889
	UL	168	16791	0.4641	0.085	0.9965	0.9824
	UI	171	17072	0.4507	0.0754	0.9966	0.9827
upa-small	OL	142	8448	0.7206	0.0211	0.7747	0.1921
	OI	141	8431	0.7202	0.0212	0.7744	0.1931
	UL	157	17613	0.3983	0.0198	0.9773	0.8947
	UI	159	17772	0.404	0.0197	0.9784	0.9012
pa-rnd	OL	108	5900	0.9232	0.5182	0.9576	0.7931
	OI	108	5900	0.9229	0.5182	0.9572	0.7931
	UL	173	17136	0.478	0.0791	1.0	1.0
	UI	159	18133	0.4277	0.0615	0.9958	0.9752
pa-small	OL	108	7532	0.8685	0.3684	0.9308	0.7542
	OI	108	7532	0.8685	0.3684	0.9308	0.7542
	UL	177	18977	0.4549	0.0778	0.9866	0.9465
	UI	183	18607	0.4503	0.0639	0.987	0.9482

Table 44: Post-processing decomposition – Dataset: d11

2.1.6 Dataset d12 – nc = 10%

	V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	100	100	7688	0.9906	0.6667
OI	100	100	7688	0.9906	0.6667
UL	1010	1010	42099	0.7054	0.0212
UI	1013	1013	41495	0.7067	0.0211

Table 45: Starting decomposition – Dataset: d12

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	122	8585	0.8998	0.3537	0.9084	0.5417
	OI	121	8583	0.9002	0.3558	0.9089	0.5455
	UL	1012	42242	0.7001	0.0174	0.9984	0.9902
	UI	1015	41638	0.7014	0.0173	0.9985	0.9902
upa-small	OL	182	10353	0.7395	0.0108	0.7413	0.0406
	OI	182	10356	0.7391	0.0108	0.7409	0.0406
	UL	850	35265	0.6969	0.0032	0.989	0.9335
	UI	844	35327	0.6844	0.0032	0.9903	0.9399
pa-rnd	OL	180	10488	0.747	0.0108	0.7499	0.0566
	OI	176	10473	0.7528	0.011	0.7557	0.0575
	UL	927	36627	0.6773	0.0029	0.9908	0.942
	UI	933	37227	0.6747	0.0029	0.9905	0.9403
pa-small	OL	175	10639	0.7502	0.0073	0.7544	0.0577
	OI	176	10644	0.749	0.0073	0.7531	0.0575
	UL	912	38642	0.6713	0.002	0.9894	0.939
	UI	917	37983	0.6515	0.002	0.9893	0.9403

Table 46: Post-processing decomposition – Dataset: d12

2.1.7 Dataset d13 – nc = 10%

V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	100	9226	0.9987	0.9417
OI	100	9226	0.9987	0.9417
UL	1128	82247	0.7724	0.014
UI	1133	80840	0.7766	0.014

Table 47: Starting decomposition – Dataset: d13

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	126	10285	0.8903	0.4581	0.8915	0.4868
	OI	126	10285	0.8925	0.4581	0.8937	0.4868
	UL	1134	82548	0.7667	0.0073	0.9971	0.9807
	UI	1140	81143	0.7707	0.0073	0.9968	0.98
upa-small	OL	187	12964	0.7449	0.0035	0.7455	0.0177
	OI	185	12961	0.7469	0.0035	0.7474	0.0179
	UL	1107	77556	0.7601	0.0008	0.9911	0.9451
	UI	1106	76668	0.7582	0.0008	0.9909	0.9451
pa-rnd	OL	192	13129	0.7396	0.0034	0.7404	0.0246
	OI	190	13124	0.743	0.0035	0.7438	0.0247
	UL	1144	79875	0.7683	0.0008	0.9919	0.9477
	UI	1146	80244	0.7625	0.0008	0.9912	0.9469
pa-small	OL	168	12964	0.7797	0.0152	0.7844	0.1213
	OI	171	12971	0.7743	0.015	0.779	0.1198
	UL	1096	80035	0.7542	0.0034	0.9922	0.9507
	UI	1097	78898	0.757	0.0034	0.9926	0.9508

Table 48: Post-processing decomposition – Dataset: d13

2.1.8 Dataset d14 – nc = 10%

	V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	100	100	14990	0.9987	0.7391
OI	100	100	14990	0.9987	0.7391
UL	1212	1212	181362	0.8313	0.0155
UI	1211	1211	179710	0.8233	0.0155

Table 49: Starting decomposition – Dataset: d14

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	114	15535	0.9364	0.4965	0.9376	0.6719
	OI	114	15535	0.9366	0.4965	0.9378	0.6719
	UL	1215	181513	0.8295	0.0123	0.9988	0.991
	UI	1214	179861	0.8215	0.0123	0.9987	0.991
upa-small	OL	199	18166	0.7426	0.0034	0.7426	0.0034
	OI	198	18163	0.7442	0.0034	0.7442	0.0034
	UL	1256	182900	0.8254	0.0007	0.9963	0.9779
	UI	1255	182693	0.8266	0.0007	0.9968	0.9787
pa-rnd	OL	193	17835	0.7502	0.0034	0.7506	0.0245
	OI	192	17831	0.7512	0.0034	0.7516	0.0246
	UL	1237	181257	0.8253	0.0007	0.9935	0.9586
	UI	1238	179010	0.8214	0.0007	0.9934	0.9586
pa-small	OL	200	17461	0.7363	0.0	0.7363	0.0
	OI	199	17459	0.7378	0.0	0.7378	0.0
	UL	1220	171392	0.8151	0.0	0.9937	0.9628
	UI	1215	170494	0.8126	0.0	0.994	0.9634

Table 50: Post-processing decomposition – Dataset: d14

2.2 Varying number of users

Dataset	nr	nu	np	$mrcu$	mpr
d21	200	500	1500	3	150
d22	200	1000	1500	3	150
d23	200	3000	1500	3	150
d24	200	5000	1500	3	150

Table 51: Constant nr and np , varying nu , and $mrcu \times mpr \approx np/3.33$

2.2.1 Dataset d21 – $nc = 8$

V	$ \mathcal{R} $	WSC	Sim_{gm}	Jacc_{gm}
OL	306	20333	0.8532	0.2778
OI	306	20218	0.8525	0.2778
UL	449	47224	0.8391	0.1019
UI	449	47093	0.8382	0.1037

Table 52: Starting decomposition – Dataset: d21

Modality	V	$ \mathcal{R} $	WSC	Sim_{gp}	Jacc_{gp}	Sim_{mp}	Jacc_{mp}
upa-rnd	OL	310	20356	0.8484	0.2655	0.9937	0.9618
	OI	310	20241	0.8478	0.2655	0.9938	0.9618
	UL	451	47233	0.8373	0.0997	0.9977	0.9868
	UI	451	47102	0.8365	0.1015	0.9979	0.9868
upa-small	OL	302	19963	0.8415	0.2463	0.9855	0.9226
	OI	301	19760	0.8407	0.2407	0.9869	0.9256
	UL	439	44186	0.8333	0.0814	0.99	0.9463
	UI	439	43960	0.8321	0.0795	0.9901	0.9463
pa-rnd	OL	298	20192	0.8681	0.2924	0.9952	0.9701
	OI	298	20183	0.8674	0.2891	0.9953	0.9701
	UL	406	39494	0.8426	0.142	0.9987	0.9926
	UI	406	39242	0.8409	0.1377	0.9987	0.9926
pa-small	OL	293	19172	0.8461	0.282	0.9804	0.8918
	OI	298	19210	0.8415	0.2784	0.9774	0.8839
	UL	439	45035	0.8336	0.1156	0.9889	0.942
	UI	437	44999	0.8348	0.114	0.991	0.9506

Table 53: Post-processing decomposition – Dataset: d21

2.2.2 Dataset d22 – nc = 8

	V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	220	17938	0.9557	0.5385	
OI	220	17927	0.9558	0.5441	
UL	749	80078	0.8236	0.0509	
UI	749	80296	0.8231	0.0498	

Table 54: Starting decomposition – Dataset: d22

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	220	17938	0.9557	0.5385	1.0	1.0
	OI	220	17927	0.9558	0.5441	1.0	1.0
	UL	749	80078	0.8236	0.0509	1.0	1.0
	UI	749	80296	0.8231	0.0498	1.0	1.0
upa-small	OL	239	18679	0.9174	0.4441	0.9793	0.9028
	OI	240	18687	0.9156	0.4426	0.9797	0.9032
	UL	794	83803	0.8223	0.0485	0.9941	0.9701
	UI	795	84015	0.8205	0.0463	0.9938	0.9701
pa-rnd	OL	234	18368	0.93	0.55	0.9849	0.9129
	OI	234	18379	0.9303	0.55	0.9826	0.9008
	UL	751	80256	0.8197	0.0485	0.9969	0.9802
	UI	751	80252	0.817	0.0485	0.9965	0.9802
pa-small	OL	238	18051	0.9182	0.5208	0.9807	0.9065
	OI	238	18049	0.9182	0.5208	0.9807	0.9065
	UL	753	80907	0.8221	0.0612	0.9935	0.9685
	UI	751	80918	0.8229	0.0626	0.994	0.9697

Table 55: Post-processing decomposition – Dataset: d22

2.2.3 Dataset d23 – nc = 8

	V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	200	200	20744	1.0	1.0
OI	200	200	20744	1.0	1.0
UL	1905	1905	214944	0.8202	0.0204
UI	1906	1906	215565	0.8178	0.0194

Table 56: Starting decomposition – Dataset: d23

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	205	20885	0.9876	0.9286	0.9876	0.9286
	OI	205	20885	0.9877	0.9286	0.9877	0.9286
	UL	1906	214979	0.8198	0.0199	0.9997	0.9984
	UI	1907	215600	0.8173	0.0189	0.9997	0.9984
upa-small	OL	209	22341	0.9743	0.8341	0.9767	0.8894
	OI	209	22375	0.9743	0.8259	0.9767	0.8894
	UL	1961	235616	0.816	0.0127	0.9978	0.9883
	UI	1962	235508	0.8166	0.0131	0.9975	0.9878
pa-rnd	OL	211	21681	0.9729	0.8026	0.9801	0.8904
	OI	211	21724	0.9731	0.7948	0.9803	0.8904
	UL	1936	223372	0.8163	0.0142	0.9996	0.9974
	UI	1937	221919	0.8164	0.0142	0.9995	0.9974
pa-small	OL	209	22011	0.972	0.6901	0.9763	0.8894
	OI	209	22009	0.972	0.6901	0.9763	0.8894
	UL	1919	230909	0.8007	0.0115	0.9979	0.9886
	UI	1921	231056	0.8019	0.0119	0.9977	0.9881

Table 57: Post-processing decomposition – Dataset: d23

2.2.4 Dataset d24 – nc = 8

	V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	200	200	25171	1.0	1.0
OI	200	200	25171	1.0	1.0
UL	3012	3012	349714	0.8173	0.0107
UI	3016	3016	351958	0.8158	0.0107

Table 58: Starting decomposition – Dataset: d24

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	203	25328	0.9928	0.9563	0.9928	0.9563
	OI	203	25328	0.9928	0.9563	0.9928	0.9563
	UL	3012	349714	0.8173	0.0107	1.0	1.0
	UI	3016	351958	0.8158	0.0107	1.0	1.0
upa-small	OL	208	27738	0.9766	0.7288	0.9777	0.8889
	OI	208	27738	0.9766	0.7288	0.9777	0.8889
	UL	3143	389212	0.8025	0.0075	0.9988	0.993
	UI	3138	393012	0.7987	0.0079	0.999	0.9933
pa-rnd	OL	208	26125	0.98	0.8133	0.9804	0.8889
	OI	208	26125	0.9798	0.8133	0.9802	0.8889
	UL	3151	371115	0.7984	0.0084	0.9997	0.9975
	UI	3153	373072	0.8008	0.0081	0.9997	0.9975
pa-small	OL	208	27650	0.9733	0.6126	0.9761	0.8889
	OI	208	27650	0.9733	0.6126	0.9761	0.8889
	UL	3017	346551	0.805	0.0078	0.9986	0.9927
	UI	3018	348137	0.8048	0.0078	0.9985	0.9924

Table 59: Post-processing decomposition – Dataset: d24

2.2.5 Dataset d21 – nc = 10%

	V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	276	17404	0.8667	0.2986	
OI	276	17488	0.8682	0.313	
UL	429	43328	0.853	0.1257	
UI	429	43259	0.8529	0.1257	

Table 60: Starting decomposition – Dataset: d21

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	286	17474	0.8525	0.2571	0.9822	0.8986
	OI	289	17579	0.8502	0.2584	0.9785	0.8709
	UL	434	43358	0.8476	0.1146	0.9939	0.9658
	UI	434	43289	0.8475	0.1146	0.9938	0.9658
upa-small	OL	351	20371	0.7956	0.1159	0.9301	0.6334
	OI	352	20345	0.7988	0.1179	0.928	0.6294
	UL	475	43686	0.8171	0.0434	0.9558	0.767
	UI	475	43447	0.8194	0.045	0.9565	0.767
pa-rnd	OL	316	21582	0.8388	0.241	0.9755	0.8498
	OI	317	21487	0.8364	0.2404	0.9745	0.8473
	UL	454	48383	0.8349	0.1049	0.988	0.929
	UI	454	48458	0.8335	0.1068	0.9878	0.929
pa-small	OL	285	18221	0.8469	0.3054	0.9605	0.7941
	OI	283	17987	0.8457	0.3071	0.9603	0.7961
	UL	445	40563	0.823	0.0898	0.9752	0.863
	UI	447	40852	0.822	0.0932	0.9727	0.8593

Table 61: Post-processing decomposition – Dataset: d21

2.2.6 Dataset d22 – nc = 10%

	V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	247	20414	0.9229	0.4752	
OI	248	20428	0.9212	0.4737	
UL	802	92314	0.8214	0.0481	
UI	802	93011	0.8235	0.0503	

Table 62: Starting decomposition – Dataset: d22

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	276	20687	0.8775	0.3408	0.9479	0.7148
	OI	274	20695	0.8808	0.3428	0.9548	0.7228
	UL	814	92405	0.8155	0.0379	0.9928	0.9564
	UI	815	93107	0.8171	0.0389	0.9922	0.9529
upa-small	OL	352	19212	0.7751	0.0203	0.813	0.1831
	OI	352	19206	0.7744	0.0203	0.8119	0.1807
	UL	812	81400	0.7954	0.007	0.9732	0.8372
	UI	814	80514	0.7933	0.008	0.9713	0.8374
pa-rnd	OL	368	19956	0.7667	0.0053	0.813	0.1712
	OI	371	19779	0.7637	0.0053	0.8083	0.166
	UL	811	83275	0.7922	0.003	0.9623	0.7872
	UI	808	82960	0.7917	0.003	0.9641	0.792
pa-small	OL	364	20099	0.7633	0.0018	0.8068	0.1562
	OI	366	20104	0.7622	0.0018	0.8045	0.1556
	UL	813	80374	0.7913	0.001	0.9702	0.8339
	UI	813	80696	0.7907	0.001	0.9696	0.8339

Table 63: Post-processing decomposition – Dataset: d22

2.2.7 Dataset d23 – nc = 10%

	V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	200	21389	0.9989	0.8519	
OI	200	21389	0.9989	0.8519	
UL	1870	213802	0.82	0.0187	
UI	1876	212655	0.8186	0.0191	

Table 64: Starting decomposition – Dataset: d23

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	240	22692	0.9133	0.4815	0.9143	0.5658
	OI	241	22694	0.9129	0.4799	0.9139	0.5638
	UL	1876	213997	0.8169	0.0157	0.9983	0.9904
	UI	1881	212848	0.8156	0.0161	0.9985	0.991
upa-small	OL	364	27585	0.7639	0.0018	0.7675	0.0424
	OI	377	27580	0.7545	0.0017	0.758	0.0414
	UL	1952	222980	0.8045	0.0005	0.9902	0.9403
	UI	1953	223566	0.8031	0.0005	0.9898	0.9403
pa-rnd	OL	382	27095	0.7476	0.0	0.7478	0.0034
	OI	383	27100	0.7469	0.0	0.7471	0.0034
	UL	1886	215631	0.8002	0.0	0.9895	0.9383
	UI	1888	212942	0.7977	0.0	0.9891	0.9368
pa-small	OL	349	27201	0.776	0.0055	0.7781	0.085
	OI	347	27194	0.7777	0.0055	0.7798	0.0853
	UL	1939	217888	0.8057	0.0014	0.9909	0.9439
	UI	1939	218685	0.8048	0.0014	0.9911	0.9448

Table 65: Post-processing decomposition – Dataset: d23

2.2.8 Dataset d24 – nc = 10%

	V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	200	200	24879	0.9995	0.9231
OI	200	200	24879	0.9995	0.9231
UL	2948	2948	336421	0.803	0.0119
UI	2944	2944	335635	0.8003	0.0119

Table 66: Starting decomposition – Dataset: d24

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	231	26436	0.9328	0.5963	0.9333	0.645
	OI	230	26434	0.9343	0.5985	0.9347	0.6475
	UL	2951	336569	0.802	0.0109	0.9996	0.997
	UI	2946	335781	0.7994	0.0109	0.9996	0.9973
upa-small	OL	365	35734	0.7594	0.0036	0.7603	0.0386
	OI	363	35737	0.76	0.0036	0.761	0.0387
	UL	2985	366644	0.7917	0.0006	0.9945	0.9685
	UI	2983	365536	0.7926	0.0006	0.9949	0.9695
pa-rnd	OL	388	36231	0.7498	0.0	0.7498	0.0
	OI	390	36236	0.7481	0.0	0.7481	0.0
	UL	3123	385338	0.7984	0.0	0.9955	0.9721
	UI	3121	382035	0.797	0.0	0.9954	0.9724
pa-small	OL	384	35932	0.747	0.0	0.747	0.0017
	OI	385	35934	0.7469	0.0	0.7469	0.0017
	UL	3077	372994	0.7954	0.0	0.9958	0.9733
	UI	3074	374515	0.7941	0.0	0.9954	0.9723

Table 67: Post-processing decomposition – Dataset: d24

2.3 Varying number of roles and users

Dataset	nr	nu	np	$mrcu$	mpr
d31	10	100	1500	3	150
d32	50	500	1500	3	150
d33	100	1000	1500	3	150
d34	500	5000	1500	3	150

Table 68: Constant np , varying nr and nu , and $mrcu \times mpr \approx np/3.33$

2.3.1 Dataset d31 – $nc = 8$

V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	10	1111	1.0	1.0
OI	10	1111	1.0	1.0
UL	51	4137	0.7597	0.0339
UI	53	4864	0.7994	0.0161

Table 69: Starting decomposition – Dataset: d31

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	11	1129	0.9523	0.75	0.9523	0.75
	OI	11	1129	0.9523	0.75	0.9523	0.75
	UL	51	4137	0.7597	0.0339	1.0	1.0
	UI	53	4864	0.7994	0.0161	1.0	1.0
upa-small	OL	18	1102	0.7708	0.0769	0.7708	0.0769
	OI	18	1102	0.7696	0.0769	0.7696	0.0769
	UL	53	3394	0.7736	0.0	0.9832	0.8909
	UI	53	3079	0.7436	0.0	0.9859	0.8909
pa-rnd	OL	18	1371	0.7736	0.0769	0.7736	0.0769
	OI	18	1371	0.7735	0.0769	0.7735	0.0769
	UL	65	4782	0.7397	0.0135	0.9922	0.9545
	UI	66	5193	0.7356	0.0	0.9923	0.9552
pa-small	OL	18	1052	0.7588	0.0769	0.7588	0.0769
	OI	18	1052	0.7587	0.0769	0.7587	0.0769
	UL	40	2486	0.7503	0.0204	0.9581	0.7907
	UI	39	2709	0.7522	0.0208	0.957	0.7857

Table 70: Post-processing decomposition – Dataset: d31

2.3.2 Dataset d32 – nc = 8

V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	50	5354	1.0	1.0
OI	50	5354	1.0	1.0
UL	370	41325	0.8129	0.0194
UI	371	41747	0.8114	0.0194

Table 71: Starting decomposition – Dataset: d32

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	51	5375	0.9902	0.9423	0.9902	0.9423
	OI	51	5375	0.99	0.9423	0.99	0.9423
	UL	370	41325	0.8129	0.0194	1.0	1.0
	UI	371	41747	0.8114	0.0194	1.0	1.0
upa-small	OL	59	5234	0.9123	0.5352	0.9215	0.6418
	OI	59	5234	0.9123	0.5352	0.9215	0.6418
	UL	359	39289	0.8024	0.0099	0.9883	0.9373
	UI	359	38899	0.801	0.0099	0.9894	0.9373
pa-rnd	OL	60	5055	0.914	0.5714	0.9316	0.6471
	OI	60	5057	0.9138	0.5714	0.9314	0.6471
	UL	313	30859	0.8256	0.0225	0.9982	0.9904
	UI	313	30820	0.8244	0.0225	0.9985	0.9904
pa-small	OL	58	5147	0.9169	0.6364	0.9169	0.6364
	OI	58	5147	0.9169	0.6364	0.9169	0.6364
	UL	335	36557	0.8106	0.0079	0.9865	0.93
	UI	334	35663	0.8018	0.0079	0.9875	0.9327

Table 72: Post-processing decomposition – Dataset: d32

2.3.3 Dataset d33 – nc = 8

	V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	100	100	10070	0.9995	0.9231
OI	100	100	10070	0.9995	0.9231
UL	680	680	76691	0.8275	0.0304
UI	682	682	76676	0.8269	0.0303

Table 73: Starting decomposition – Dataset: d33

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	104	10150	0.9794	0.8214	0.9799	0.8889
	OI	104	10150	0.9795	0.8214	0.9799	0.8889
	UL	683	76740	0.8251	0.0262	0.9977	0.9869
	UI	685	76725	0.8245	0.0261	0.9979	0.9869
upa-small	OL	109	9826	0.9476	0.7712	0.9517	0.7949
	OI	109	9826	0.9476	0.7712	0.9517	0.7949
	UL	651	71590	0.8084	0.0218	0.9925	0.9651
	UI	651	70160	0.8092	0.0204	0.9924	0.9651
pa-rnd	OL	111	10144	0.9502	0.6614	0.9628	0.7983
	OI	111	10148	0.9504	0.6614	0.9629	0.7983
	UL	668	73353	0.8276	0.0254	0.9983	0.9881
	UI	670	73146	0.8253	0.0253	0.9984	0.9881
pa-small	OL	108	9208	0.9555	0.7931	0.9555	0.7931
	OI	108	9208	0.9555	0.7931	0.9555	0.7931
	UL	675	67816	0.8145	0.0238	0.9933	0.9663
	UI	675	67429	0.8114	0.0238	0.9933	0.9663

Table 74: Post-processing decomposition – Dataset: d33

2.3.4 Dataset d34 – nc = 8

	V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	512	512	51071	0.9864	0.6063
OI	512	512	51032	0.9864	0.6063
UL	3388	3388	419578	0.7973	0.0256
UI	3388	3388	419019	0.796	0.0256

Table 75: Starting decomposition – Dataset: d34

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	514	51128	0.9843	0.5994	0.9979	0.9884
	OI	514	51089	0.9842	0.5994	0.9978	0.9884
	UL	3389	419610	0.7969	0.0253	0.9998	0.9991
	UI	3389	419051	0.7957	0.0253	0.9998	0.9991
upa-small	OL	518	51765	0.9795	0.7023	0.9898	0.9544
	OI	518	51753	0.9795	0.7023	0.9898	0.9544
	UL	3500	451602	0.7936	0.0233	0.9986	0.9934
	UI	3500	450800	0.7935	0.0233	0.9986	0.9934
pa-rnd	OL	510	52226	0.9871	0.5562	0.9962	0.9767
	OI	510	52226	0.9871	0.5562	0.9961	0.9767
	UL	3434	442657	0.7915	0.0247	1.0	1.0
	UI	3436	438579	0.7916	0.025	1.0	1.0
pa-small	OL	518	48715	0.9784	0.6967	0.989	0.9544
	OI	518	48714	0.9784	0.6967	0.989	0.9544
	UL	3385	410866	0.785	0.024	0.9986	0.9938
	UI	3390	408828	0.7863	0.0237	0.9984	0.9932

Table 76: Post-processing decomposition – Dataset: d34

2.3.5 Dataset d31 – nc = 10%

V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	10	979	1.0	1.0
OI	10	979	1.0	1.0
UL	53	3182	0.7245	0.0328
UI	53	3206	0.7183	0.0328

Table 77: Starting decomposition – Dataset: d31

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	11	992	0.9538	0.75	0.9538	0.75
	OI	11	992	0.9538	0.75	0.9538	0.75
	UL	53	3182	0.7245	0.0328	1.0	1.0
	UI	53	3206	0.7183	0.0328	1.0	1.0
upa-small	OL	20	1252	0.7342	0.0	0.7342	0.0
	OI	20	1252	0.7342	0.0	0.7342	0.0
	UL	52	3449	0.715	0.0	0.9609	0.7857
	UI	49	3779	0.7609	0.0	0.9562	0.7736
pa-rnd	OL	11	889	0.9537	0.75	0.9537	0.75
	OI	11	889	0.9537	0.75	0.9537	0.75
	UL	43	2504	0.7225	0.0392	1.0	1.0
	UI	43	2383	0.7333	0.0392	1.0	1.0
pa-small	OL	11	967	0.9474	0.75	0.9474	0.75
	OI	11	967	0.9474	0.75	0.9474	0.75
	UL	51	3568	0.7568	0.0167	0.9888	0.9423
	UI	51	3542	0.7383	0.0167	0.9888	0.9423

Table 78: Post-processing decomposition – Dataset: d31

2.3.6 Dataset d32 – nc = 10%

	V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	50	5004	1.0	1.0	1.0
OI	50	5004	1.0	1.0	1.0
UL	314	35039	0.8472	0.043	0.043
UI	314	34992	0.84	0.043	0.043

Table 79: Starting decomposition – Dataset: d32

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	60	5197	0.9129	0.5714	0.9129	0.5714
	OI	60	5197	0.9134	0.5714	0.9134	0.5714
	UL	317	35109	0.8404	0.0309	0.9949	0.9657
	UI	318	35064	0.8324	0.0308	0.9935	0.9627
upa-small	OL	100	6172	0.7399	0.0	0.7508	0.0201
	OI	101	6174	0.7391	0.0	0.7499	0.02
	UL	341	36005	0.8053	0.0	0.9857	0.9145
	UI	341	36109	0.8047	0.0	0.9853	0.9145
pa-rnd	OL	99	5802	0.7394	0.0	0.7394	0.0
	OI	99	5802	0.7398	0.0	0.7398	0.0
	UL	339	33650	0.8086	0.0	0.9854	0.914
	UI	339	33509	0.802	0.0	0.985	0.914
pa-small	OL	100	6042	0.7402	0.0	0.7452	0.0134
	OI	100	6042	0.7397	0.0	0.7447	0.0134
	UL	355	35790	0.8123	0.0	0.9895	0.9366
	UI	354	35767	0.8149	0.0	0.9901	0.9365

Table 80: Post-processing decomposition – Dataset: d32

2.3.7 Dataset d33 – nc = 10%

	V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	100	100	9742	1.0	1.0
OI	100	100	9742	1.0	1.0
UL	688	688	76229	0.8281	0.0287
UI	688	688	75763	0.8238	0.0287

Table 81: Starting decomposition – Dataset: d33

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	121	10190	0.9122	0.5563	0.9122	0.5563
	OI	121	10190	0.9132	0.5563	0.9132	0.5563
	UL	691	76292	0.8254	0.0246	0.9979	0.987
	UI	691	75825	0.8211	0.0246	0.9978	0.987
upa-small	OL	188	11600	0.7526	0.0035	0.7614	0.043
	OI	192	11611	0.7494	0.0034	0.7588	0.0424
	UL	705	70812	0.8093	0.0012	0.9861	0.9213
	UI	703	69942	0.8048	0.0012	0.9873	0.9265
pa-rnd	OL	193	11330	0.7434	0.0	0.7492	0.0208
	OI	195	11330	0.741	0.0	0.7467	0.0206
	UL	682	70757	0.8041	0.0	0.984	0.9091
	UI	683	70334	0.8014	0.0	0.983	0.9078
pa-small	OL	199	11970	0.739	0.0	0.744	0.0135
	OI	199	11970	0.7389	0.0	0.7442	0.0135
	UL	700	77821	0.8074	0.0	0.9835	0.9047
	UI	701	76511	0.8024	0.0	0.9833	0.9021

Table 82: Post-processing decomposition – Dataset: d33

2.3.8 Dataset d34 – nc = 10%

	V	$ \mathcal{R} $	WSC	Sim _{gm}	Jacc _{gm}
OL	510	510	51108	0.9878	0.5906
OI	510	510	51105	0.9878	0.5906
UL	3391	3391	414766	0.7927	0.0248
UI	3394	3394	414037	0.795	0.0245

Table 83: Starting decomposition – Dataset: d34

Modality	V	$ \mathcal{R} $	WSC	Sim _{gp}	Jacc _{gp}	Sim _{mp}	Jacc _{mp}
upa-rnd	OL	573	52390	0.9338	0.4081	0.9452	0.7002
	OI	571	52383	0.9354	0.4092	0.9465	0.7024
	UL	3402	415011	0.7905	0.0215	0.9983	0.9897
	UI	3404	414281	0.793	0.0212	0.9984	0.99
upa-small	OL	871	59186	0.7789	0.0111	0.7829	0.0683
	OI	871	59193	0.7787	0.0111	0.7831	0.0683
	UL	3494	426710	0.774	0.0018	0.9888	0.9302
	UI	3499	424850	0.7723	0.0018	0.9884	0.9297
pa-rnd	OL	838	59028	0.7879	0.003	0.7977	0.1223
	OI	834	59047	0.7889	0.003	0.7985	0.1227
	UL	3480	421564	0.7819	0.001	0.9864	0.9152
	UI	3480	419152	0.783	0.001	0.986	0.9155
pa-small	OL	829	59636	0.7906	0.0023	0.7981	0.1058
	OI	840	59686	0.7865	0.0022	0.7945	0.1057
	UL	3492	426726	0.7724	0.0008	0.9888	0.9321
	UI	3504	423732	0.7756	0.0007	0.9885	0.9288

Table 84: Post-processing decomposition – Dataset: d34

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